

Sequences

Section 9.1

2-17-26

A) Review

$$\{a_n\}$$

Ex:

$$\left\{\frac{1}{n}\right\} = \frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \dots$$

$$f(x) = \frac{1}{x}$$


$$a_n = \frac{1}{n}$$

Ex: 3^n

$$n=0 \quad 3^0, 3^1, 3^2, 3^3, \dots$$

$$n=1 \quad 3^1, 3^2, 3^3, 3^4, \dots$$

Ex:

$$\frac{2}{n!} \quad n=0 \quad \frac{2}{0!}, \frac{2}{1!}, \frac{2}{2!}, \frac{2}{3!}, \frac{2}{4!}, \dots$$

B) Write a general term

$$\{3, 8, 13, 18, \dots\}$$

↳ Arithmetic

$$n=0 \quad a_n = 5n + 3$$

$$n=1 \quad a_n = 5n - 2$$

$$\{5, -15, 45, -135, \dots\}$$

↳ Geometric

$$n=0 \quad a_n = 5(-3)^n$$

$$n=1 \quad a_n = 5(-3)^{n-1}$$

$$\{0, 3, 8, 15, 24, \dots\}$$

↳ Quadratic

$$n=1 \quad a_n = n^2 - 1$$

C) Even, Odd, Alternating

Even	Odd	Alternating
$2n$	$2n-1$	n
$2n+2$	$2n+1$	$n+1$
$2n-2$		

D) Does it converge?

$$\text{Ex: } \left\{\frac{1}{n}\right\} \quad \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Converges to 0

$$\text{Ex: } \left\{\frac{(-1)^n}{n}\right\} \quad \lim_{n \rightarrow \infty} \frac{(-1)^n}{n}$$

Converges to 0

$$\text{Ex: } \left\{\frac{3n+4}{5n-2}\right\} \quad \lim_{n \rightarrow \infty} \left(\frac{3n+4}{5n-2}\right) = \frac{3}{5}$$

Converges to $\frac{3}{5}$

$$\text{Ex: } \left\{\left(\frac{3}{2}\right)^n\right\} \quad \lim_{n \rightarrow \infty} \left(\frac{3}{2}\right)^n = \infty$$

Diverges

$$\text{Ex: } \left\{\frac{(n+1)!}{(n+3)!}\right\} \quad \lim_{n \rightarrow \infty} \frac{(n+1)!}{(n+3)!}$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)(n)(n-1)\dots}{(n+3)(n+2)(n+1)(n)(n-1)\dots}$$

$$= 0 \quad \text{Converges to 0}$$

$$\text{Ex: } \left\{(7n)^{1/2}\right\} \quad \lim_{n \rightarrow \infty} (7n)^{1/2}$$

∞ no natural log to L'H and get ∞